SOVEREIGN EDUCATIONAL INFRASTRUCTURE

RENANCE

Nigeria's Sovereign Dual-Engine Educational Operating System

Eliminating Examination Panic for Candidates • Ending Paper & Administrative Drain for Schools

PILLAR I: Institutional School OS

- 1-Tap NERDC National Curriculum auto-installation (Primary 1 - SSS 3)
- Ink-Saving single-topic B&W lesson notes (Slashing paper costs by 70%+)
- 30-Second mobile attendance roll call, gate kiosk & conflict-free timetables
- Automated continuous assessment (CA 20/20/60) & student rank computation
- Kobo-safe integer fee ledger with live debtor roster & serialized ID cards

PILLAR II: Cognitive Student Study OS

- 207,172+ verified questions across 681 packs (JAMB, WAEC, NECO, Post-UTME, Uni)
- Proprietary CBT Exam Clock Coach & post-exam Panic Forensics autopsy
- Everyday Relatable Scenarios translating abstract STEM formulas into Nigerian reality
- Socratic AI Tutor providing hint-first guided inquiry without spoon-feeding
- 1v1 Multiplayer CBT Arena with WebSocket live sync & Spaced Repetition Queue

207,172+
VERIFIED PAST QUESTIONS

681 PACKS
1978 - 2025 COVERAGE

774 LGAs
SOVEREIGN ROLLOUT SURFACE

100% OFFLINE
ZERO-DATA ANDROID SQLITE

THE NATIONAL CRISIS

The Twin Bottlenecks Crippling Nigerian Education

Why Millions Fail High-Stakes Exams While School Administrators Drown in Paper Chaos

The Student Crisis: Clock Panic & Rote Recall

- 1.4M+ UTME Candidates Fail Cutoffs: Over 75% of applicants fail to secure tertiary admission annually; 76% struggle in WAEC science and mathematics.
- The Exam Hall Clock Trap: Candidates freeze on complex early questions, spending 4-5 minutes on a single mark. Acute panic sets in, working memory collapses, and 10-15 marks are lost to frantic blind guessing.
- Abstract Textbook Alienation: Imported Western pedagogy encourages rote memorization. Students recite formulas verbatim without physical comprehension.
- The Bandwidth Barrier: High telecommunications tariffs price out 85% of Nigerian candidates who cannot stream heavy 40-minute educational videos.

The School Crisis: Administrative & Financial Hemorrhage

- Astronomical Paper & Toner Drain: Schools spend over NGN 300,000 every term on paper registers, handwritten lesson plan notebooks, and terminal exam photocopies.
- Chaotic Fee Accounting & Debtors: Bursars rely on paper ledgers or loose desktop spreadsheets. Parents exploit poor records, causing millions in uncollected fee arrears.
- Rampant Report Card Forgery: Razor-blade alterations and NGN 500 cyber-cafe counterfeit report sheets destroy academic credibility and parent trust.
- Teacher Administrative Burnout: Exhausting late-night lesson plan copying robs educators of teaching energy, while manual CA calculation breeds human error.

KEY INSIGHT: Commercial 'quiz dump' apps fail because they test without teaching; video portals fail because data is unaffordable. Nigeria requires an infrastructure-level Operating System.

ARCHITECTURAL INNOVATION

A Sovereign Dual-Engine Operating System

Unifying Institutional School Governance and Cognitive Student Learning into One Sovereign Platform

Pillar I: Institutional School OS

[+] 1-Tap NERDC Installation: Pre-populates Primary 1 to SSS 3 curriculum across Science, Arts, Commercial & Technical.

[+] Ink-Saving B&W Print Engine: Strict single-topic lesson note exports cut paper/toner costs by 70%+.

[+] 30-Second Morning Roll Call: Intuitive mobile roster and entrance gate kiosk mode with SUBEB CSV exports.

[+] Conflict-Free Timetable: Atomic commit prevents classroom or teacher scheduling collisions.

[+] CA 20/20/60 & Public Checker: Automated grading + 6-digit scratch PIN at /school/check.

[+] Kobo-Integer Fee Ledger: Exact accounting eliminates rounding loss; live debtor tracking prevents evasion.

Pillar II: Cognitive Student OS

[+] 207k+ Verified Questions: Curated past papers across 681 packs spanning JAMB, WAEC, NECO, and Tertiary.

[+] CBT Exam Clock Coach: Real-time pacing gauge (ON TRACK, LINGERING, TRAP RISK) eliminates hall panic.

[+] Post-Exam Panic Forensics: Pinpoints exact time-traps and calculates marks lost to clock mismanagement.

[+] Everyday Relatable Scenarios: Explains Newton's laws through Danfo buses, and osmosis through honey beans.

[+] Socratic AI Tutor: Guided hints without spoon-feeding correct letters; deterministic offline matrices.

[+] 1v1 CBT Multiplayer Arena: High-energy 5-min academic duels, leaderboards, and spaced review queue.

Unified Cross-Platform Surfaces

[+] Web Launcher (Next.js 15): Over 1,446 pre-rendered static routes deployed on global edge networks with sub-second loading.

[+] Mobile Shell (Flutter Android APK): Ultra-lightweight APK packing embedded SQLite; 5 dedicated Offline Desks.

[+] Zero-Data Operation: Download exam packs once via Wi-Fi; practice 100% offline in remote villages without airtime.

[+] Desktop Gate Kiosk: Plug-and-play attendance and gate verification on inexpensive Windows/Linux laptops.

[+] ADR-0003 Anti-Cheat Security: Sealed answer keys kept strictly server-side; sha256 checksums verify bundle integrity.

CURRICULUM VAULT

Unrivaled National Question Repository: 207,172+ Banked

681 Specialized Academic Packs Spanning 1978 to 2025 Across Secondary, Screening, and Tertiary Levels

JAMB UTME GATEWAY

43,553 Verified Questions | 46 Subject Packs

- Full 4-Subject Official Configuration (Use of English + 3 electives)
- Realistic 2-hour official countdown clock with question navigator
- Complete coverage: Sciences, Social Sciences, Arts, and Commercial tracks
- Full comprehension passages, poetry analysis, and worked step-by-step models

WAEC WASSCE & NECO SSCE

40,077 Questions | 72 Comprehensive Series

- WAEC (1988-2025): 37,696 questions covering Objective and Theory papers
- NECO SSCE & GCE: 2,381 verified curriculum-aligned questions
- Worked theory solutions showing step-by-step marking scheme rubrics
- Covers Mathematics, English, Physics, Chemistry, Biology, Economics, Lit-in-English

POST-UTME SCREENING

Premier Nigerian Universities Screening Series

- Authentic screening papers for University of Lagos (UNILAG)
- Federal University of Technology, Akure (FUTA) & UI screening series
- Obafemi Awolowo University (OAU) & University of Benin (UNIBEN)
- Aptitude tests, general paper screening, and speed drills for 200+ cutoffs

TERTIARY UNIVERSITY DESKS

122,000+ Questions | 548 Course Modules

- Foundational 100 & 200 Level Nigerian University course examination banks
- AMS 101 Principles of Management, COS 101 Computer Programming
- MEE 101 Engineering Mechanics, General Studies (GST 101/102)
- Semester lecture notes, past exam pools, and continuous CGPA forecasting

PROPRIETARY COGNITIVE ENGINE

The CBT Exam Clock Coach: Neutralizing Hall Panic

Dynamic Real-Time Pacing Guidance and Post-Exam Clock Forensics That Recover 10–15 Lost Marks

Dynamic CBT Pacing Engine: How It Works

The Mathematical Target Dwell Time (T_target):

T_target = Total Exam Duration / Total Number of Questions

(In a 40-question, 40-minute paper, baseline is exactly 60 seconds per item).

Live Ambient Pacing Badge (Non-Intrusive Subconscious Conditioning):

[ON TRACK - GREEN] (t <= 1.2 x T_target): Answering within steady, optimal cadence.

[LINGERING - AMBER] (1.2 < t <= 2.0 x T_target): Subtle warning that safety buffer is eroding.

[TIME TRAP RISK - RED] (t > 2.0 x T_target): Visual circuit breaker prompting candidate to guess, flag, and preserve available marks ahead.

Psychological Conditioning: Candidates instinctively internalize target velocity, eliminating the sudden 30-minute panic shock common in JAMB centers.

Post-Exam Panic Forensics Autopsy

Temporal Reconstruction of the Entire Paper:

Compares candidate's actual speed curve against ideal linear velocity.

Time-Trap Isolation:

Specifically identifies questions where a student sacrificed 3 to 5 minutes on a single item yet still chose an incorrect option.

Panic Rush Zone Detection:

Exposes late-exam questions answered in under 8 seconds due to expiring timers, proving where working memory collapsed.

Calculated Lost Marks Metric:

Empowers students by showing: 'You lost 8 marks to clock panic, not intellectual inability.' Transforming practice into scientific calibration.

PEDAGOGICAL INNOVATION

Everyday Relatable Scenarios & Socratic AI

Translating Abstract Foreign Theorems into Lived Nigerian Realities and Guided Critical Thinking

Everyday Relatable Scenarios Engine

>> Newton's First Law (Inertia)
Standing inside a speeding yellow Danfo bus on Ikorodu Road when the driver abruptly hits the brakes - your upper torso lurches forward because your mass resists change in velocity.

>> Newton's Third Law (Action/Reaction)
Stepping off a wooden canoe onto the muddy shore of a Lagos lagoon - the forward push of your foot exerts an equal and opposite force that kicks the canoe backward into the water.

>> Osmosis & Semi-Permeable Membranes
Soaking rock-hard, wrinkled dried honey beans in fresh water overnight - discovering them swollen and soft by morning as water permeates the bean skin to dilute concentrated starches.

>> Opportunity Cost & Scarcity (Economics)
Holding only NGN 1,500 and deciding between an internet data bundle for school assignments or a hot plate of cafeteria jollof rice - the true cost is the meal forfeited.

Socratic AI Tutor (Hint-First Pedagogy)

[+] Anti-Spoon-Feeding Law: Refuses to reveal multiple-choice letters (A, B, C, D) or copy-paste answers, preventing intellectual dependency.

[+] Diagnostic Guided Inquiry: When a student misses a question, the AI asks probing questions to isolate the core rule, variable, or grammar distinction.

[+] Step-by-Step Walking: Guides the candidate through equations (e.g. 'Look at Kinetic Energy Ek = 1/2mv^2. If speed doubles, what happens to velocity squared?').

[+] Deterministic Offline Hint Matrices: If internet connectivity drops, the system falls back to pre-compiled rule matrices stored inside local SQLite.

[+] Curriculum-Grounded: Trained exclusively on verified NERDC, JAMB, and WAEC syllabi, eliminating AI hallucinations.

RETENTION & GAMIFICATION

1v1 Multiplayer CBT Arena & Spaced Repetition

Turning Solitary Exam Drills into an Electrifying Social Sport While Halting Cognitive Decay

1v1 Multiplayer CBT Arena (/arena)

[*] Electrifying 5-Minute Academic Duels: Two candidates anywhere in Nigeria enter a virtual ring for a 15-question rapid-fire clash drawn from their shared syllabus.

[*] Ultra-Low Latency WebSocket Sync: Live synchronized score counters tick upward with sub-100ms response times, consuming negligible data bytes.

[*] Automated Study Bot Sparring: If no human peer matches within 20 seconds at odd hours, an intelligent study bot matched to candidate skill steps in.

[*] Gamified XP & Daily Streaks: Duels earn Experience Points (XP), level up avatars, and maintain consecutive study streaks.

[*] National School Leaderboards: Weekly state and school rankings foster healthy academic rivalry and pride.

Spaced-Repetition Review Queue (/review)

[*] Automated Mistake Harvesting: Every incorrect or skipped question from CBT mocks or practice sessions is instantly queued for retention.

[*] Ebbinghaus Forgetting Curve Calibration: Vulnerabilities resurface at scientifically calculated intervals: 24 Hours, 72 Hours, and 7 Days later.

[*] Targeted Remediation vs Blind Cramming: Eliminates the waste of re-sitting entire 50-question papers; targets only documented cognitive gaps.

[*] Permanent Memory Consolidation: Questions are graduated from the queue only after two consecutive correct answers under timed pressure.

[*] Seamless Sync: Fully synchronized between online web sessions and offline Android SQLite databases.

TECHNICAL INFRASTRUCTURE

Zero-Data Sovereign Architecture & 5 Offline Desks

Engineered to Flourish Where Mobile Broadband Fails and Electrical Power Fluctuates Across Nigeria

The 5 Offline Desks

[Desk] 1. JAMB UTME Offline Desk: Full 46 subject packs with official 180-question mock generator.

[Desk] 2. WAEC WASSCE Offline Desk: 61 packs with objective and worked theory solutions.

[Desk] 3. NECO SSCE Offline Desk: 11 series covering standard secondary school testing.

[Desk] 4. Post-UTME University Desk: Direct screening questions for UNILAG, FUTA, UI, OAU.

[Desk] 5. School Workspace Desk: Complete teacher lesson notes, schemes, and question banks.

Android SQLite Shell

[Mobile] Native Flutter Performance: Runs smoothly even on low-cost entry-level Android smartphones.

[Mobile] Embedded SQLite Database: Stores tens of thousands of questions and syllabi locally on the device.

[Mobile] One-Time 5-Minute Sync: Connect to Wi-Fi once, download packs, and study for 6 months completely offline.

[Mobile] Zero Data Consumption: Full CBT mock exams, Clock Coach, and everyday scenarios function in Airplane Mode.

[Mobile] Standalone APK Distribution: Direct download without requiring high-friction app store accounts.

Go Backend & Edge Web

[Cloud] Go 1.27 Study API: Sub-15ms execution latency using goroutine worker pools for async grading.

[Cloud] Next.js 15 Static Generation: Over 1,446 pre-rendered static routes on zero-cost edge networks.

[Cloud] ADR-0003 Sealed Answer Keys: Questions contain only stems and choices; keys live strictly server-side.

[Cloud] Cryptographic Bundle Checksums: SHA-256 integrity verification prevents client-side question tampering.

[Cloud] Multi-Tenant PostgreSQL: Strict separation between student study data and institutional school records.

INSTITUTIONAL EFFICIENCY

NERDC Curriculum Auto-Install & 70% Paper Slashing

Standardizing Academic Operations in 60 Seconds While Eradicating Crushing Printing Expenses

1-Tap NERDC Statutory Curriculum Installation

[+] Zero Manual Data Entry: Upon registration, Renance auto-provisions the official NERDC curriculum framework in 60 seconds.

[+] Complete National Ladder: Pre-populates Primary 1 to Primary 6, JSS 1 to JSS 3, and SSS 1 to SSS 3.

[+] All Secondary Streams: Covers Science, Arts, Commercial, and Technical subject pathways aligned with WAEC/NECO.

[+] 12-Week Schemes of Work: Instant term schemes pre-fill every subject slot. Teachers customize rather than staring at blank pages.

[+] Multi-Tier Role Isolation: Proprietors hold sovereign administrative/financial keys; teachers access only assigned classes.

Ink-Saving B&W Print Architecture (70%+ Savings)

[+] Strict Black-and-White Standard: Decorative color backgrounds and fills are banned, optimizing output for office laser printers.

[+] Single-Topic Print Optimization: Exports clean, 1-topic lesson sheets, ending wasteful 50-page notebook bulk printing.

[+] Economic Impact on School Budgets: In an era where a paper ream costs NGN 9,000 and toner cartridges cost small fortunes, Renance slashes paper expenditure from NGN 300,000+ to under NGN 90,000 per term.

[+] NERDC School Exam Pool: 1-click terminal examination generation pulls balanced, randomized papers ready for instant printing.

DAILY OPERATIONS

Attendance Roster, Gate Kiosk & Collision-Free Timetable

Streamlining Morning Roll Call, Gate Entry, and Weekly Scheduling into Synchronized Workflows

30-Second Attendance & Gate Kiosk Mode

[*] 30-Second Classroom Roll Call: Mobile tap-toggle roster eliminates 15 minutes of wasted morning roll calls.

[*] Gate Kiosk Mode: Tablet or laptop mounted at the school gate logs arrivals in real time, alerting management to truancy.

[*] SUBEB Regulatory Compliance: One-click export of monthly attendance registers in CSV/JSON format for state inspections.

[*] PTA Attendance Analytics: Generates termly student attendance percentages printed directly on verified terminal report cards.

Collision-Free Timetable Matrix

[*] Visual Weekly Matrix Builder: Intuitive grid mapping subjects, academic periods, assemblies, and recreation breaks.

[*] Atomic Single-Transaction Commit: Validates entire weekly schedule at once, preventing double-booked classrooms or overlapping teachers.

[*] Instant Staff Distribution: Once approved by management, timetable instantly propagates to all teacher personal mobile dashboards.

[*] Offline Classroom Reference: Teachers view assigned weekly schedules directly on their phones without paper printouts.

ACADEMIC INTEGRITY

Tamper-Proof Continuous Assessment & Result Checker

Eradicating Report Card Forgery and Eliminating Human Error from Terminal Grade Computation

Universal Continuous Assessment Model

[+] Nigerian Statutory Formula: CA1 (20%) + CA2 (20%) + Terminal Exam (60%) = 100%.

[+] Automated Calculations: Instantly calculates student totals, subject averages, grade boundaries (A1 to F9), and class rankings.

[+] Classroom Isolation: Teachers enter scores only for their assigned class and subject, preventing unauthorized grade manipulation.

[+] Lock & Certify Workflow: Management reviews and signs off on results; finalized records lock against retroactive tampering.

Public Parent Result Checker (/school/check)

[*] 6-Digit Cryptographic Scratch PIN: Unique session-scoped PIN generated for each enrolled pupil at term end.

[*] Zero-Travel Smartphone Access: Parents navigate to /school/check, enter Student ID + PIN, and unlock verified digital report cards.

[*] Complete Forgery Eradication: Halts razor-blade grade alterations, correction fluid tampering, and NGN 500 counterfeit cyber-cafe prints.

[*] Official Institutional Seals: Displays verified school logo, principal endorsement, subject remarks, and tamper-proof security hash.

FINANCIAL GOVERNANCE

Integer-Safe Kobo Accounting & Serialized ID Cards

Halting Revenue Leakage, Tracking Tuition Debtors, and Automating Student Identity Verification

Integer-Safe Kobo Accounting Ledger

[Naira] Exact Integer Accounting: Calculates every transaction in integer kobo (100 kobo = NGN 1.00), preventing floating-point rounding loss.

[Naira] Multi-Tier Fee Structures: Configures tuition, lab levies, uniforms, and PTA dues universally or by specific class cohorts.

[Naira] Multi-Channel Digital Receipts: Logs cash, verified bank transfer, POS, and drafts with immutable transaction receipt hashes.

[Naira] Real-Time Debtor Defaulter Roster: Principals view searchable rosters of unpaid fees, preventing exam-week fee evasion.

Automated Serialized ID Card Engine

[ID] National Matriculation Serials: Standardized institutional format: REN-2026-NNNNNN (e.g., REN-2026-000142).

[ID] Print-Ready 8-Card Batch Sheets: Exports printable sheets with pupil photographs, blood groups, emergency contacts, barcodes, and seals.

[ID] Bypasses External Graphic Vendors: Schools print directly on office cardstock and laminate in-house, saving NGN 150,000+ yearly.

[ID] Instant Anti-Fraud Revocation: Stolen or lost credentials are blacklisted instantly without deleting historical academic records.

MARKET OPPORTUNITY

Massive National Footprint Across 774 LGAs

Capturing Nigeria's Expanding EdTech Frontier with an Unfair Offline Distribution Advantage

Total Addressable Market (TAM)

NGN 120B+

- 1.4M+ Annual JAMB UTME candidates seeking university admission.
- 1.6M+ Annual WAEC WASSCE candidates writing secondary school leaver exams.
- 1.2M+ Annual NECO SSCE candidates across public and private centers.
- Millions of basic education pupils in Primary 1 to JSS 3 requiring curriculum support.

Serviceable Market (SAM)

40,000+ Schools

- Primary and secondary institutions across 36 States + FCT seeking digital transformation.
- 250+ Tertiary Institutions (Universities, Polytechnics, Colleges of Education).
- 774 Local Government Areas: One sovereign deployment surface capable of operating offline.
- UBEC & SUBEB Partnerships: Direct institutional compliance with national digital mandates.

Teacher Income & Job Creation

Two-Sided Scale

- Earning Platform for Nigeria's Teachers: Qualified educators monetize revision notes, test pools & live lessons.
- Monetizing Real Demand: Tutors reach candidates across all 774 LGAs without physical travel.
- Unfair Distribution Advantage: Zero-data offline capability unlocks deep rural markets inaccessible to competitors.
- Low-Cost Android First: Works on inexpensive smartphones, eliminating expensive hardware requirements.

BUSINESS MODEL

Dual-Engine Monetization & High-Margin Unit Economics

Predictable High-Volume Consumer Subscriptions Paired with Recurring Institutional B2B SaaS

Student Consumer Economy (B2C)

- Free Frictionless Onboarding: 1 free full CBT simulation + 20 questions/session.
- NGN 500 / 1 Week Pass: Perfect for rapid pre-exam cramming sprints.
- NGN 1,500 / 1 Month Pass: Full access to all 681 packs and Socratic AI tutor.
- NGN 2,500 / 3 Months Pass: Termly prep for WAEC, NECO, or JAMB.
- NGN 5,000 / 1 Year All-Access (Best Value): Unrestricted web & mobile offline desks.
- REN Coin Economy: Win duels (+1 REN$), refer peers (+3 REN$), redeem 250 REN$ for 1 month premium - driving viral organic growth.

Institutional School SaaS Tiers (B2B)

- Basic (NGN 3,000/mo or NGN 10,000/yr): Up to 100 students. View-only notes, curriculum, attendance & results.
- Standard (NGN 6,000/mo or NGN 25,000/yr) - Most Popular: Up to 300 students. Exam bank, timetable, fee ledger, and ID cards.
- Premium (NGN 20,000/mo or NGN 60,000/yr): Unlimited students. Full PDF downloads, complete operations + 10 free annual student passes.
- Proprietor Friendly: School pays a flat predictable fee. Zero per-teacher or per-parent micro-charges.
- High Retention: Once curriculum, attendance, and ledgers are established, annual churn is under 5%.

HIGH-MARGIN UNIT ECONOMICS: OVER 85% GROSS MARGINS

Static Site Generation on edge CDNs combined with local SQLite on Android means marginal server cost per active offline student approaches NGN 0.00. Every additional subscriber drops straight to the bottom line.

THE BUILDERS & THE ASK

The Engineering Team, Capital Allocation & Vision

A Lean Technical Team Shipping Sovereign Educational Infrastructure for 774 Local Government Areas

The Builders: Lean Technical Leadership

[Leadership] Chief Technology Officer (CTO)
Focus: Core Go Architecture & Renance AI Training
Owns the high-throughput Go grading engine (<15ms), PostgreSQL schemas, and pedagogical AI models.

[Leadership] Lead Cybersecurity Engineer
Focus: Device-Lock & Anti-Cheat Doctrine
Owns 1-device hardware binding, ADR-0003 sealed answer keys, result encryption, and anti-fraud PINs.

[Leadership] Chief Product Engineer
Focus: Cross-Platform UX/UI & Mobile SQLite Shell
Owns the Next.js Web Launcher, Flutter Android APK, and offline-first desk performance.

Funding Utilization (The Ask)

40% - Awareness Campaigns & Proprietor Workshops
Direct roadshows onboarding 500+ private and public schools across primary education clusters.

35% - AI Token Reserves & Cloud Storage Reserves
High-throughput inference tokens for Socratic AI and multi-region database backups.

25% - Ecosystem Developer Accounts & Distribution
Apple Developer ($99), Google Play ($25), Microsoft Partner ($19), and APK distribution mirrors.

Modernizing Nigerian Education.

One sovereign platform. Two engines. 774 Local Government Areas.

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